TEACHER GUIDE · GRADE 1 · BLOCK CODING

Change One Block

Quick facilitation notes & answer key

Goal

Students alter one block of a known program at a time, find the new result, and describe how that single change affected the outcome. This is exactly C3.2 — read and alter existing code and describe how changes to the code affect the outcomes — with a predict step on the last task.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 so students can run the base code, then re-run after each one-block change.

Run it (about 20 min)

1. I DO: run the base code to 4. 'Now I change ONE block and see what happens.'

2. WE DO: Exercise 1 — forward 2 → forward 3 makes 5 (one higher, one more step). Exercise 2 — last block → back 1 makes 2 (lower, Bit moves the other way).

3. YOU DO: Exercise 3 — start at 1 → start at 0; predict 3, then run; the whole code shifts down by one.

4. Connect: 'Each change moves Bit in a way you can describe — that is reading and altering code.'

Answer key (modelled by the run-gate)

Base — start 1 → forward 2 → 3 → forward 1 → 4.

Ex 1 — forward 2 → forward 3: 1 → 4 → 5. Bit lands on 5, one HIGHER than 4 (one extra forward step).

Ex 2 — last block → back 1: 1 → 3 → 2. Bit lands on 2, LOWER, because a back block moves Bit the other way instead of forward.

Ex 3 — start at 0: 0 → 2 → 3. Bit lands on 3 — one lower than the base at every step because the whole code starts one number earlier.

Watch for & differentiate

Common slip: changing more than one block — remind students to return to the base code before each exercise (change only one thing).

Simplify: keep the base code on the floor path and physically swap one card at a time.

Extend: ask which single change to the base code would make Bit land on 1 (e.g. change forward 2 to back 2, or start at 1 → back blocks).

Success indicator: the child finds 5, 2, and 3 and explains each result in terms of the one block that changed.